Universitatea POLITEHNICA din București
Facultatea de Automatică și Calculatoare

Electronică analogică

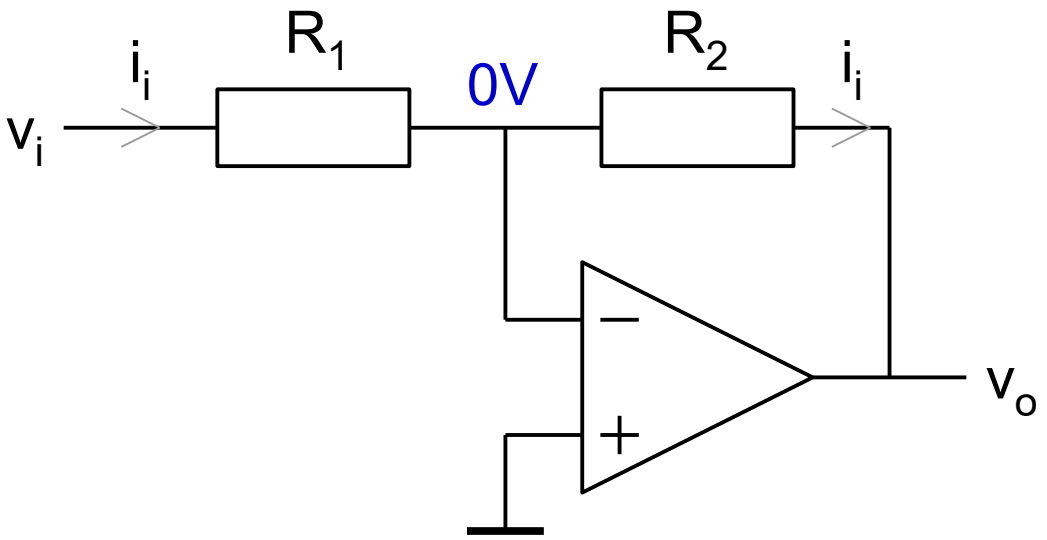
Curs – Material suplimentar 7

Convertoare logaritmice și exponențiale

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Inversor



$$i_i = \frac{v_i}{R_1}$$

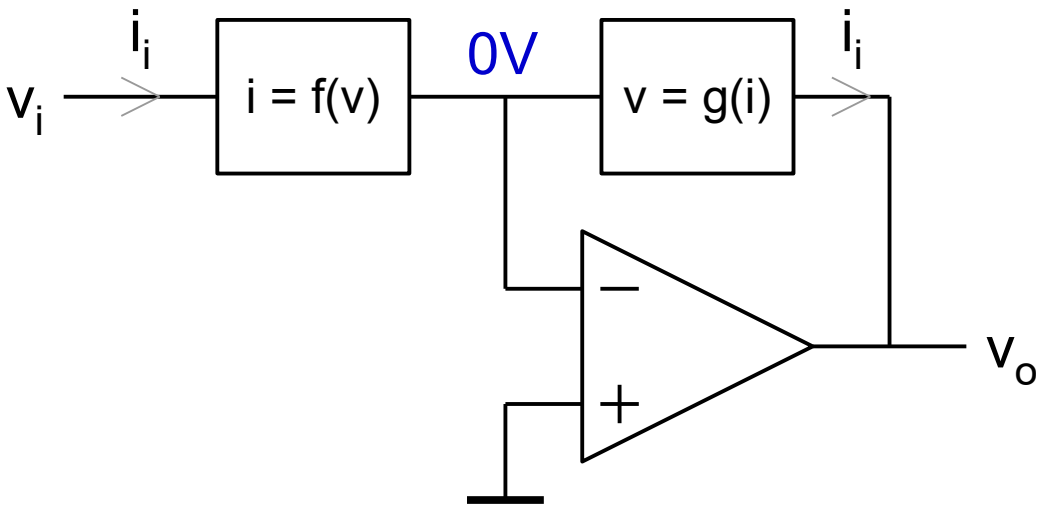
tensiune => curent

$$v_o = -i_i R_2$$

curent => tensiune

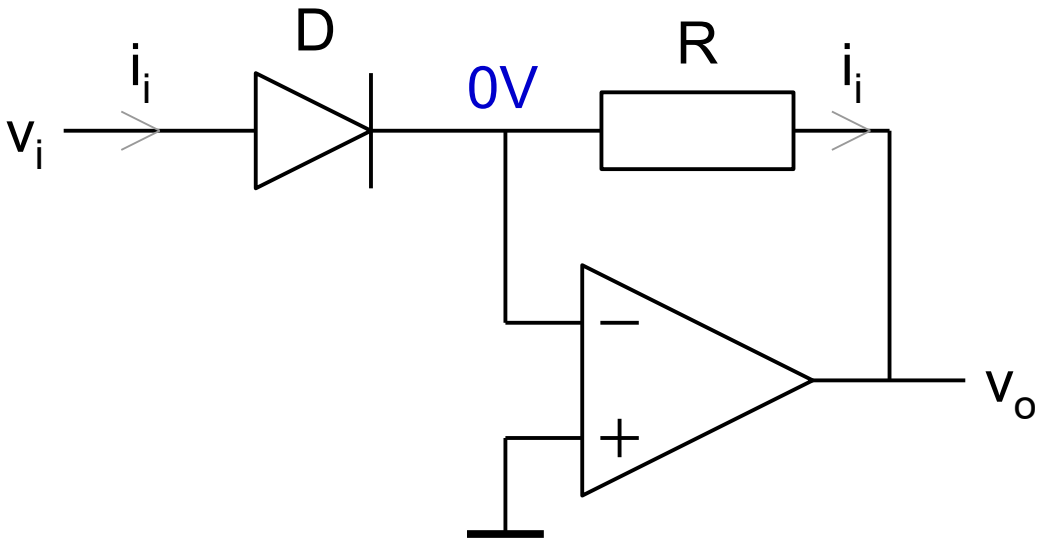
$$v_o = -v_i \frac{R_2}{R_1}$$

Inversor – generalizare



$$v_o = -g\left(f\left(v_i\right)\right)$$

Convertor exponențial



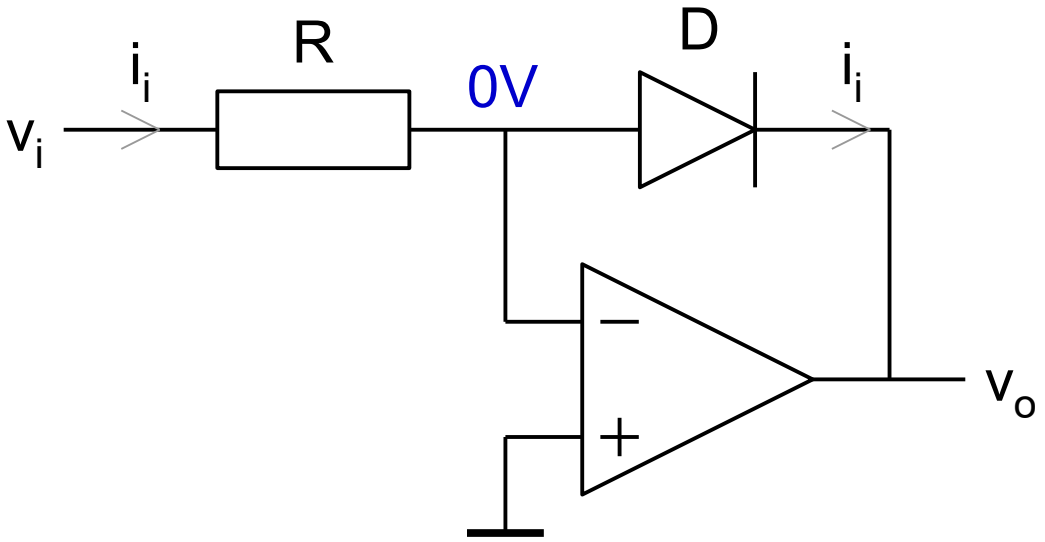
$$i_i \approx I_0 \, e^{\frac{v_i}{v_T}}$$

$$v_o = -i_i R$$

$$v_o = -R I_0 \, e^{\frac{v_i}{v_T}}$$

condiție: $v_i > 0$

Convertor logaritmic



$$i_i = \frac{v_i}{R_1} \approx I_0 \, e^{\frac{v_D}{v_T}}$$

$$v_o = -v_D$$

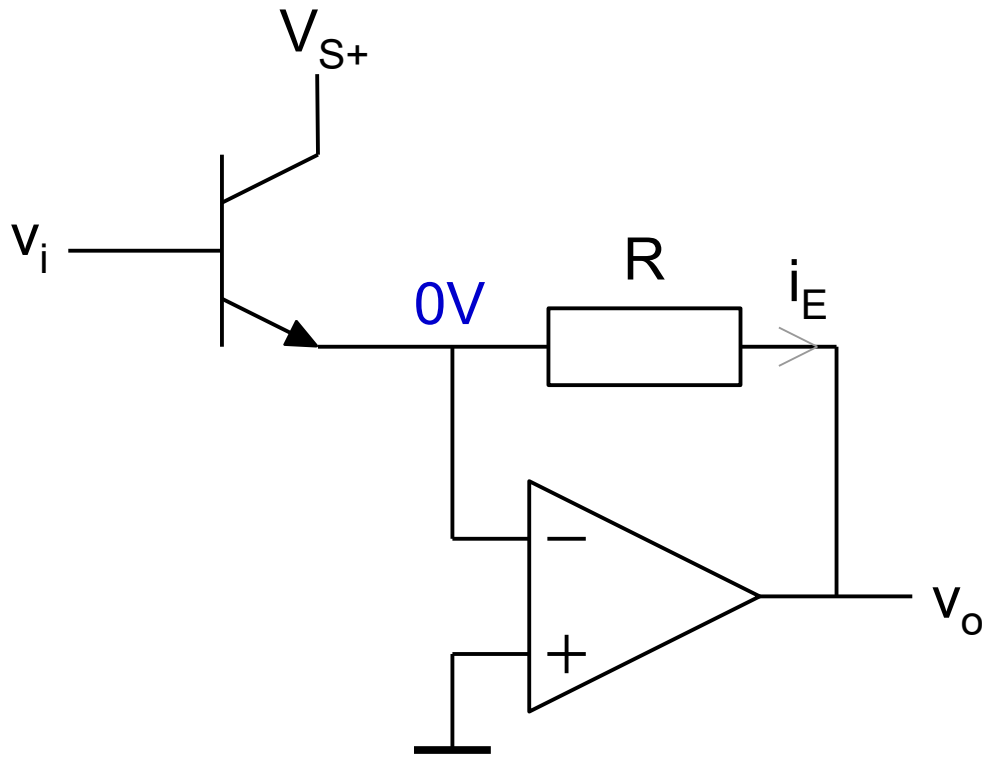
condiție: $v_i > 0$

$$v_o = -v_T \ln \frac{i_i}{I_0} = -v_T \ln \frac{v_i}{R\, I_0}$$

Observații

- se obișnuiește utilizarea de tranzistoare în locul diodelor – ecuație similară
- parametrul I_0
  - inexact specificat
  - dispersie mare
  - dependență de temperatură
- dar în realizare integrată
  - 2 tranzistoare “identice” au aprox. același I_0
  - circuite elaborate – realizează compensarea I_0

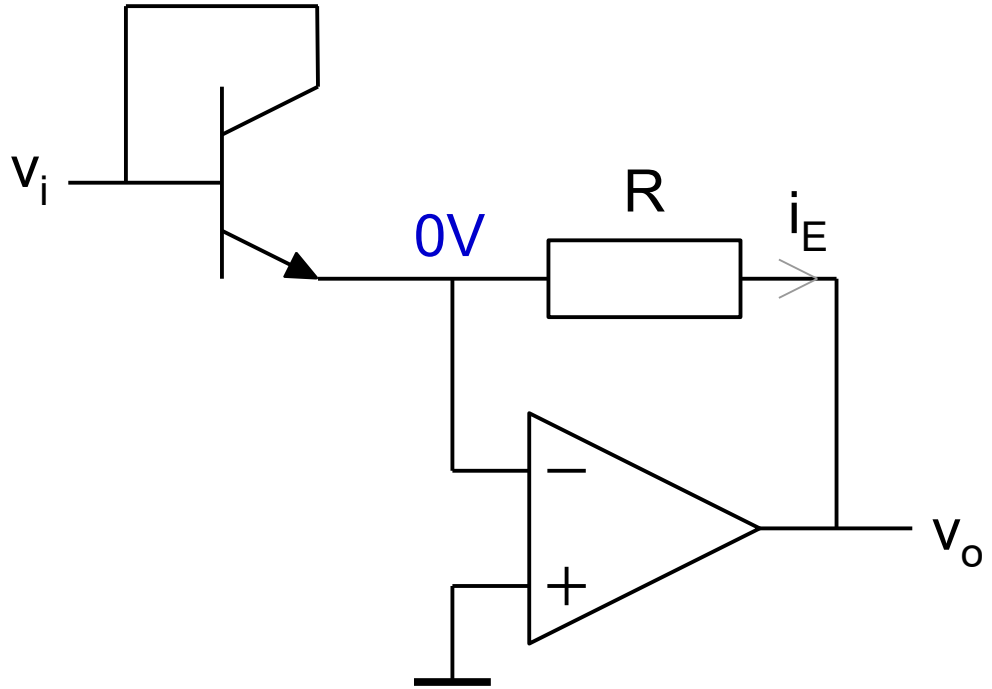
Convertor exponențial elaborat (1)



$$v_o \approx -R\, I_{E0}\, e^{\frac{v_i}{v_T}}$$

curent de intrare mai mic

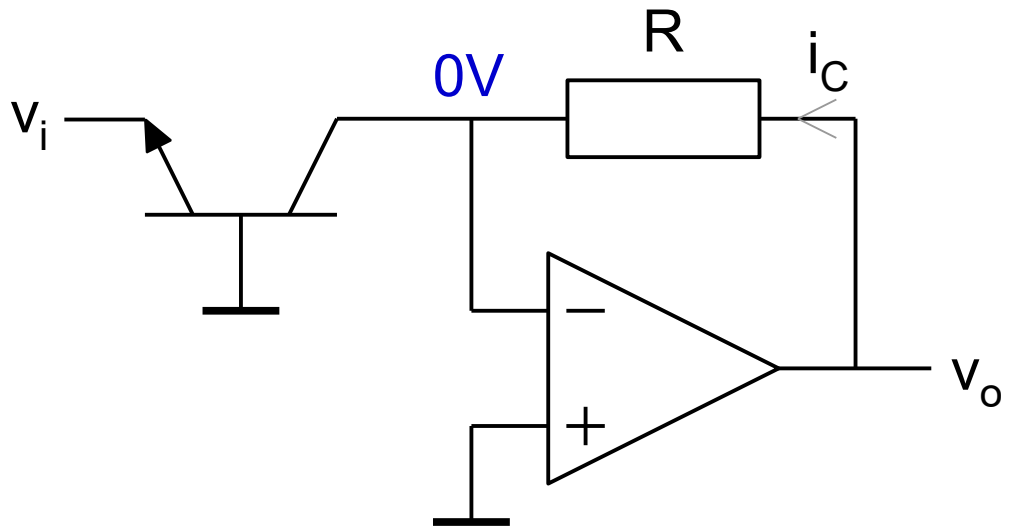
Convertor exponențial elaborat (2)



$$v_o \approx -R\, I_{E0}\, e^{\frac{v_i}{v_T}}$$

redusă influența v_{CE}

Convertor exponențial elaborat (3)

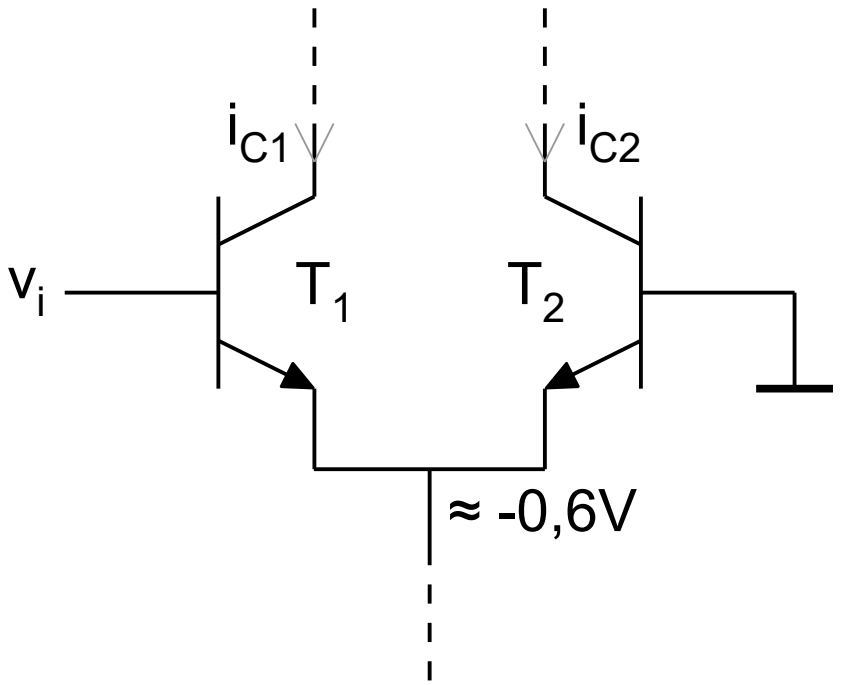


$$v_o \approx -R\,\alpha\,I_{E0}\,e^{-\frac{v_i}{v_T}}$$

variantă - redusă influența v_{CE}

$$v_i < 0$$

Convertor exponențial elaborat (4)



$$i_{C1} \approx \alpha_1 I_{E01} e^{\frac{v_{BE1}}{v_T}}$$

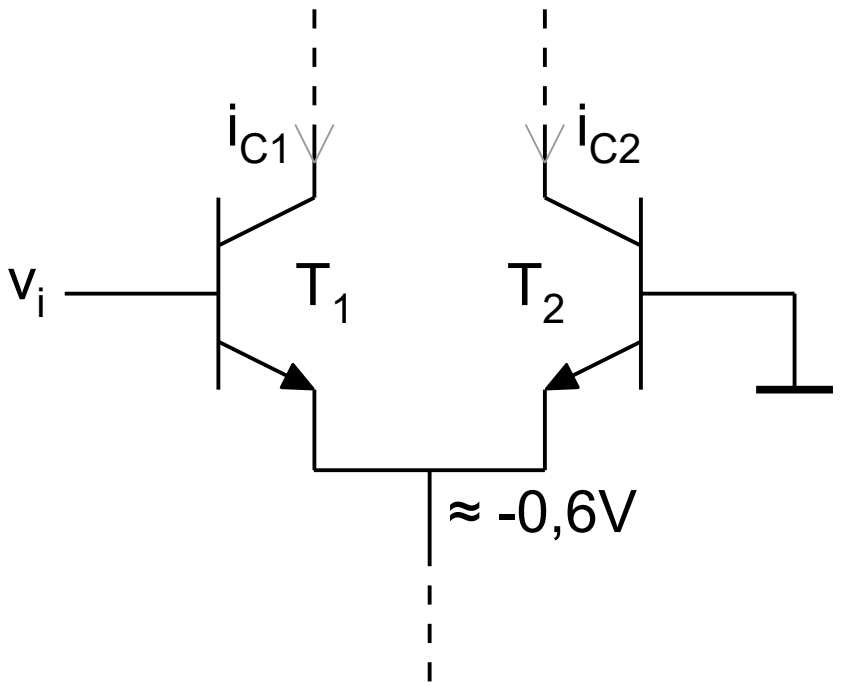
$$i_{C2} \approx \alpha_2 I_{E02} e^{\frac{v_{BE2}}{v_T}}$$

$$\frac{i_{C1}}{i_{C2}} \approx \frac{\alpha_1 I_{E01}}{\alpha_2 I_{E02}} e^{\frac{v_{BE1}-v_{BE2}}{v_T}}$$

$$\frac{\alpha_1 I_{E01}}{\alpha_2 I_{E02}} \approx 1$$

$$\frac{i_{C1}}{i_{C2}} \approx e^{\frac{v_i}{v_T}}$$

Convertor exponențial elaborat (5)



$$\frac{i_{C1}}{i_{C2}} \approx e^{\frac{v_i}{v_T}}$$

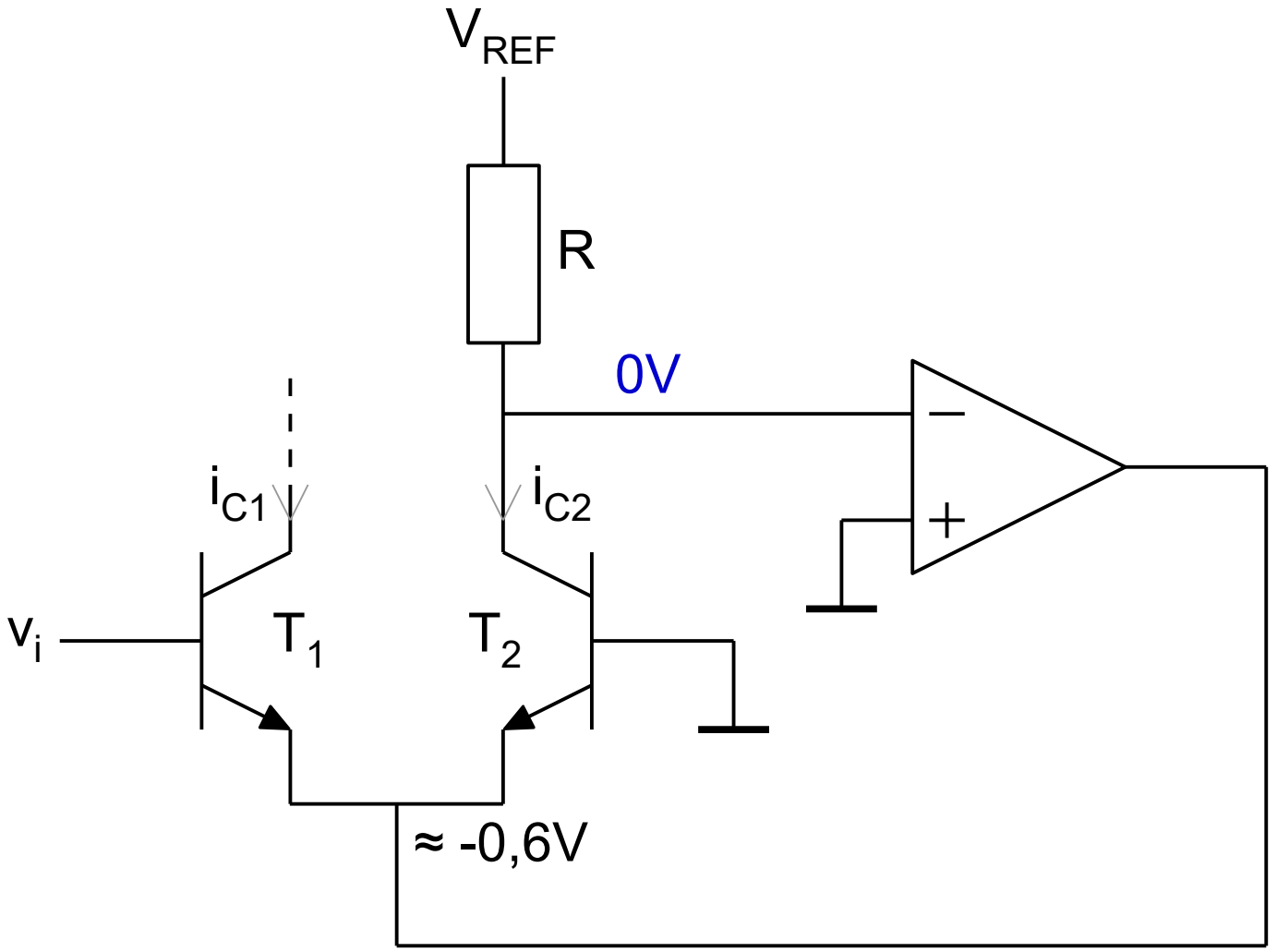
idee: AO care menține

$$i_{C2} = \text{const}$$

$$v_{C2} = 0$$

și asigură și polarizarea tranzistoarelor

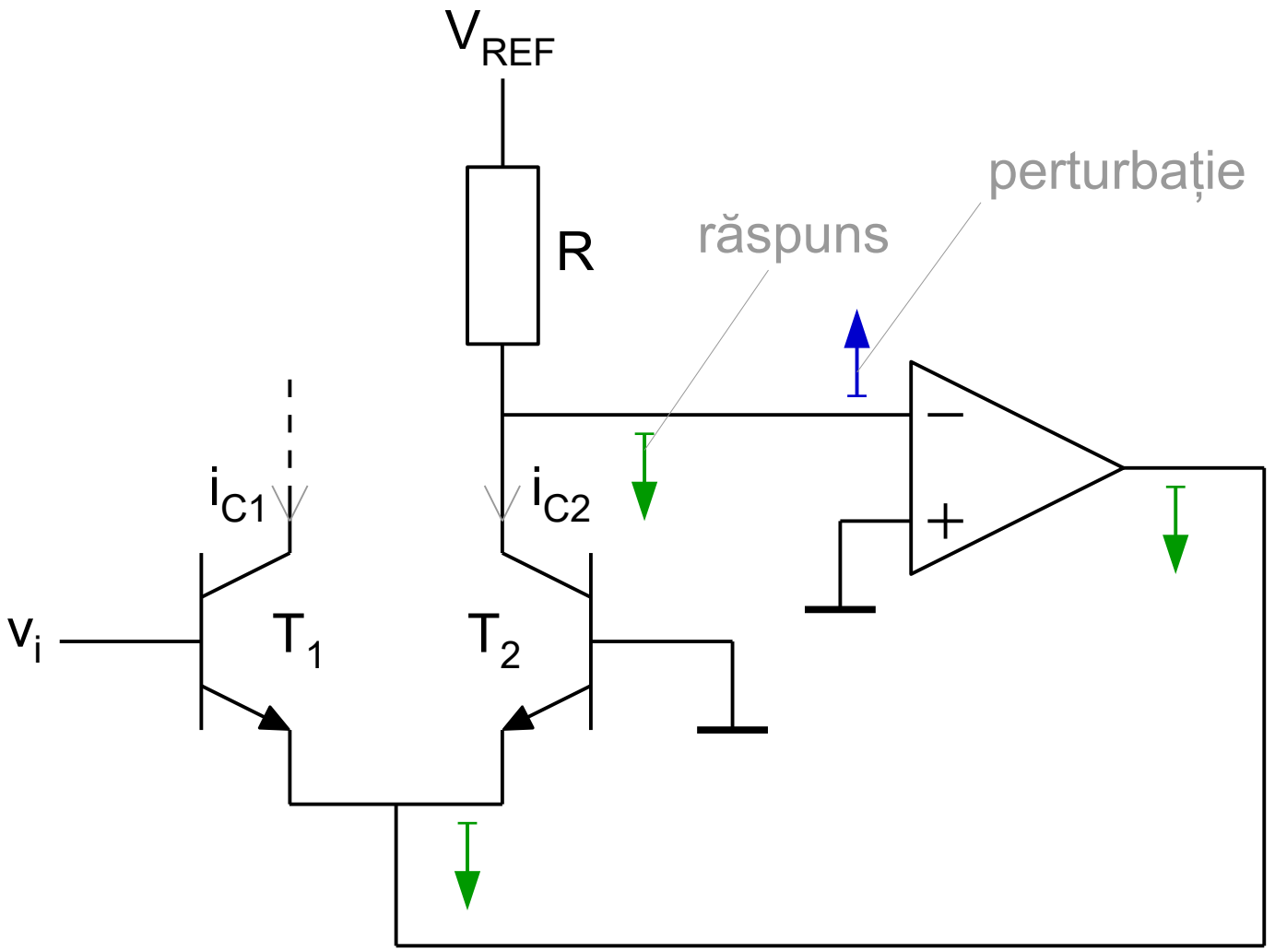
Convertor exponențial elaborat (6)



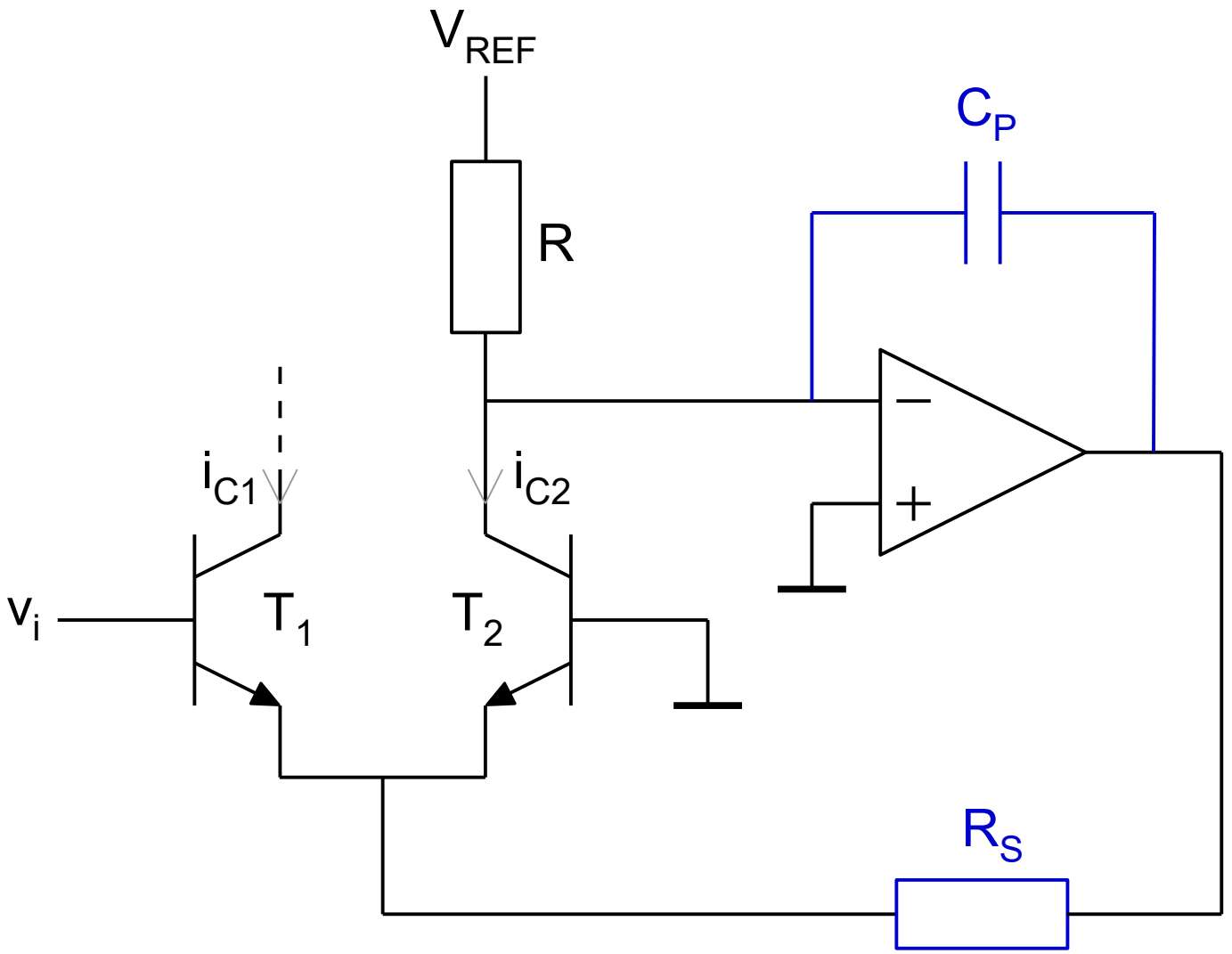
$$i_{C1} \approx I_{C2} e^{\frac{v_i}{v_T}}$$

$$I_{C2} = \frac{V_{REF}}{R}$$

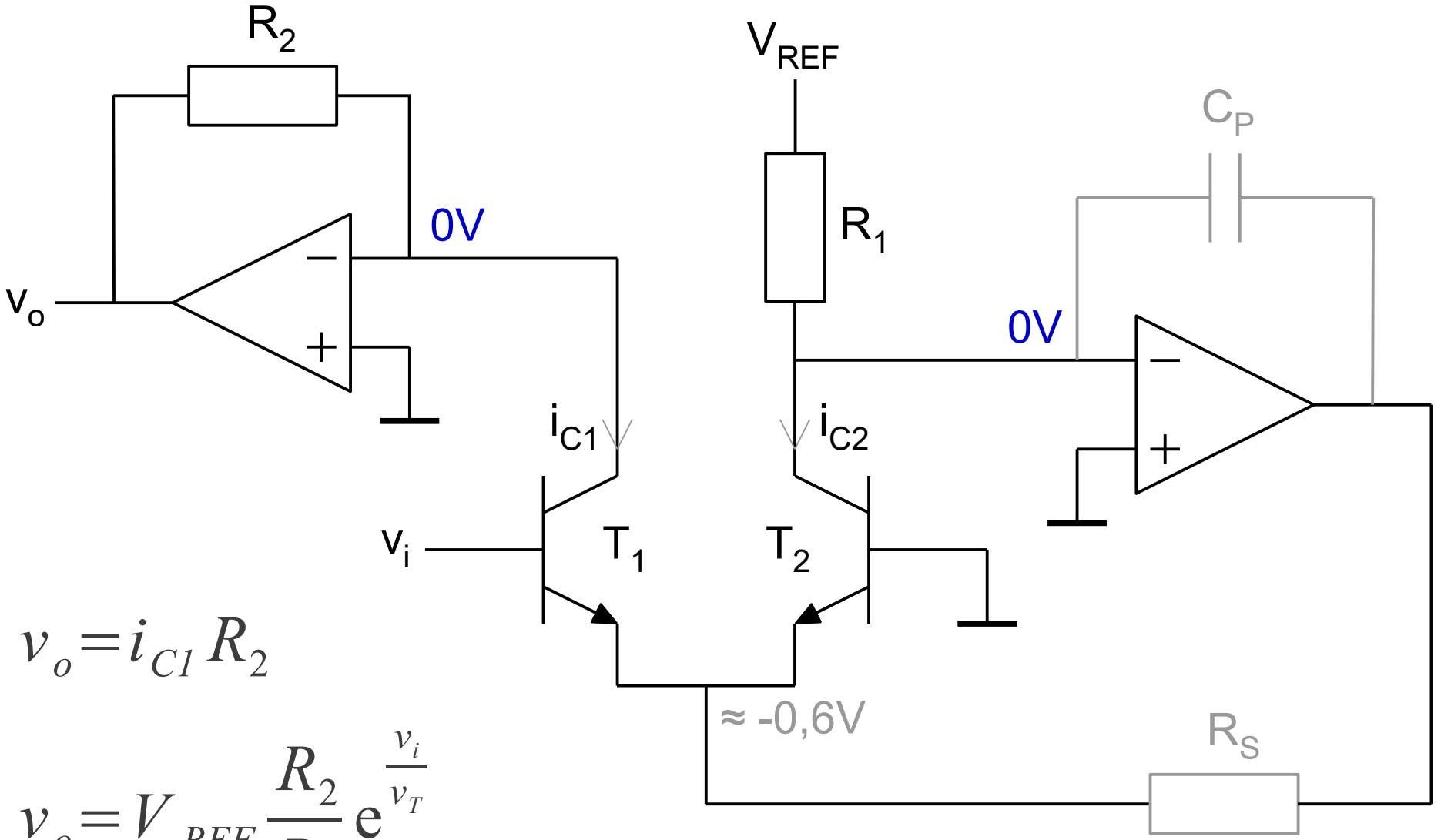
Verificare – reacție negativă



Considerații de stabilitate



Convertor exponențial elaborat (7)



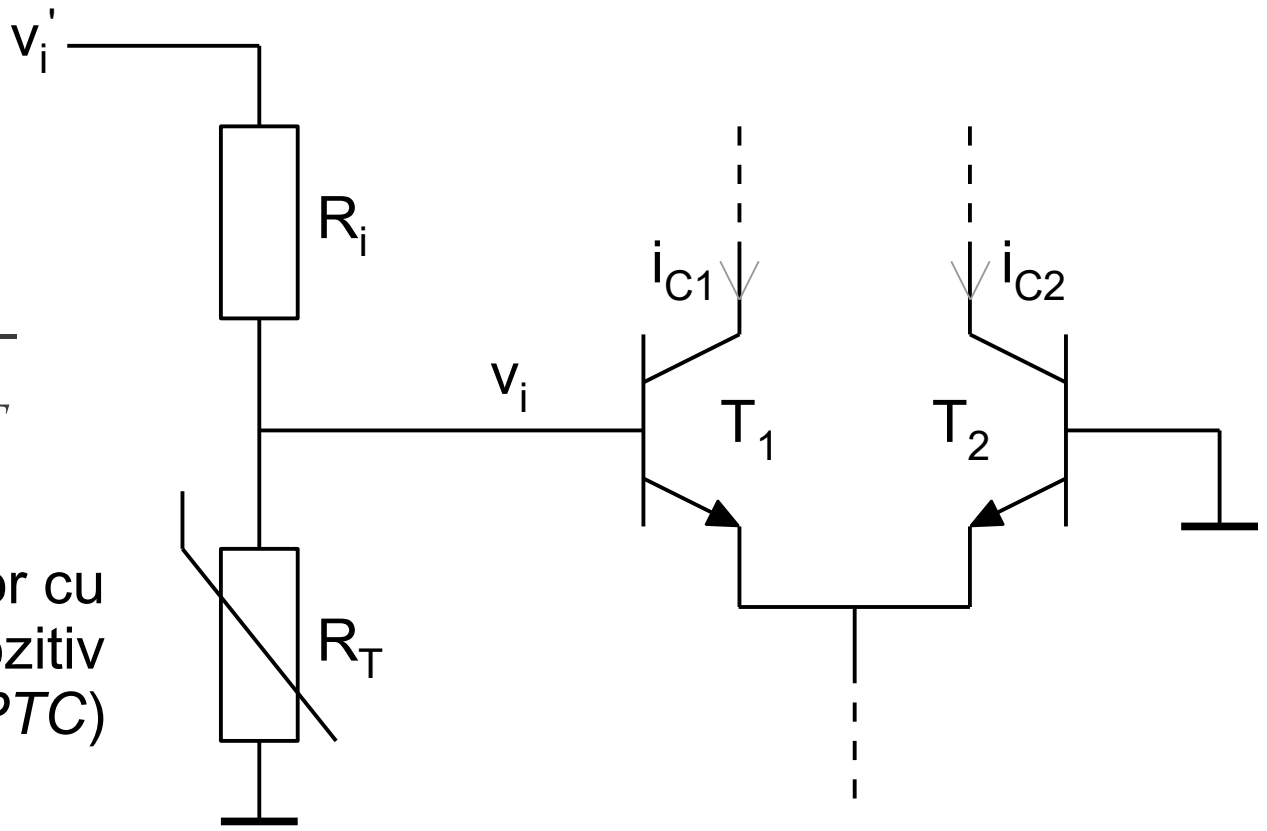
$$v_o = i_{C1} R_2$$

$$v_o = V_{REF} \frac{R_2}{R_1} e^{\frac{v_i}{v_T}}$$

Compensare termică

$$v_o = V_{REF} \frac{R_2}{R_1} e^{\frac{v_i}{v_T}} \leftarrow \frac{kT}{q}$$

$$v_i = v_i' \frac{R_T}{R_i + R_T}$$



termistor cu coeficient pozitiv (*PTC*)

Alte aplicații

- Multiplicator

$$e^{\ln a + \ln b} = a\,b$$

- ridicare la pătrat
- radical etc.

- pentru mai multe detalii:
  - documentație circuit integrat **LOG100**